

Introduction to Logic

Assignment 2

King Mongkut's Institute of Technology Ladkrabang

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Problem 1

Determine whether each sentence below can be translated into a formula in Propositional Logic using the given propositional letters and their specified meanings. If so, provide a formula that has the closest meaning to the sentence; otherwise, state that there is no translation.

Example. "If you have not paid your tuition fee, you will not be allowed to graduate."

p = You have paid your tuition fee.

g = You are allowed to graduate.

Ans. Yes. $\neg p \rightarrow \neg g$

(a) "Sweden and Norway will both not adopt the Euro."

s = Sweden will adopt the Euro.

n = Norway will adopt the Euro.

→ BOTH Sweden AND Norway will NOT adopt the Euro

$$\neg s \wedge \neg n$$

s	n	$\neg s$	$\neg n$	$\neg s \wedge \neg n$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

(b) "Sweden and Norway will not both adopt the Euro."

s = Sweden will adopt the Euro.

n = Norway will adopt the Euro.

→ Sweden or Norway may adopt the Euro, but NOT BOTH

$$\neg(s \wedge n) \equiv \neg s \vee \neg n$$

DeMorgan's

s	n	$s \wedge n$	$\neg(s \wedge n)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

(c) "Our leader doesn't ^{NOT} dye his hair, ^{OR} use makeup, or wear a wig."

h = Our leader dyes his hairs.

m = Our leader uses makeup.

w = Our leader wears a wig.

$$\neg(h \vee m \vee w) \equiv \neg h \wedge \neg m \wedge \neg w$$

DeMorgan's

(d) "By ^{IF} signing this document, ^{THEN} you agree to the terms and conditions of this software."

s = You sign this document.

a = You agree to the terms and conditions of this software.

$$s \rightarrow a$$

DeMorgan's law

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

- (e) “Unless ^{IF NOT} I see it with my own eyes, ^{AND} and hear it with my own ears, ^{THEN} I ^{NOT} never will believe it.” (Charles Dicken)

s = I see it with my own eyes.
 h = I hear it with my own ears.
 b = I believe it.

$$\neg(s \wedge h) \rightarrow \neg b$$

Implication

- (f) “The message was sent from an unknown system ^{AND} but it was not scanned for viruses.”
 u = The message was sent from an unknown system.
 s = The message was scanned for virus.

$$u \wedge \neg s$$

- (g) “Access is granted ^{IF} whenever (the user has paid the subscription fee and enters a valid password.)”

a = Access is granted.
 f = The user has paid the subscription fee.
 p = The user enters a valid password.

$$f \wedge p \rightarrow a$$

- (h) “John has a belief that both Mary and Tom lied.”

j = John has a belief. Impossible to explain what John's belief is about
 m = Mary lied.
 t = Tom lied.
 no translation

- (i) “Being affiliated with a major political party is not sufficient for you to become the President of the United States.” ^{negation of sufficiency cannot be written}
 a = You are affiliated with a major political party. ^{$\neg(a \rightarrow p)$ is only true when a is true and p is false. The meaning is changed.}
 p = You are becoming the President of the United States. no translation

- (j) “High public debt and a sharp rise in consumer prices are ^{IF AND ONLY IF} necessary and sufficient conditions for an economic crises to happen in the country.”

d = The country has high public debt.
 p = There is a sharp rise in consumer prices in the country.
 c = There is going to be an economic crisis in the country.

$$d \wedge p \leftrightarrow c$$

↪ biconditional

- (k) “John and Mary are friends.”

j = John is a friend. Impossible to write the relationship.
 m = Mary is a friend.

↓
 writing $j \wedge m$ changes the meaning. NO translation

Problem 2

Suppose SE-Rocks is a popular rock band at KMITL, whose members are the following students in the Software Engineering program: Alex, Beth, and Carl. Let p_1, p_2, p_3, q_1, q_2 and q_3 be the following propositions:

p_1 : Alex is a lead singer.

p_2 : Beth is a lead singer.

p_3 : Carl is a lead singer.

q_1 : Alex plays guitar.

q_2 : Beth plays guitar.

q_3 : Carl plays guitar.

Write the following propositions about the band using $p_1, p_2, p_3, q_1, q_2, q_3$ and logical connectives.

- (a) Beth does **not** play guitar **and** Carl is **not** a lead singer. $\neg q_2 \wedge \neg p_3$
 $\rightarrow$ Beth is NOT lead singer AND Carl is NOT lead singer
- (b) **Neither** Beth **nor** Carl is a lead singer. $\neg p_2 \wedge \neg p_3 \equiv \neg (p_2 \vee p_3)$ by DeMorgan's
 $\rightarrow$ IF lead singer THEN plays guitar
- (c) The band's lead singers **also** play guitar. $(p_1 \rightarrow q_1) \wedge (p_2 \rightarrow q_2) \wedge (p_3 \rightarrow q_3)$
- (d) There is **one** lead singer in the band. $(\underline{p_1} \wedge \neg p_2 \wedge \neg p_3) \vee (\neg p_1 \wedge \underline{p_2} \wedge \neg p_3) \vee (\neg p_1 \wedge \neg p_2 \wedge \underline{p_3})$
 EXACTLY ONE
- (e) **At least two** members of the band play guitar. $(q_1 \wedge q_2) \vee (q_1 \wedge q_3) \vee (q_2 \wedge q_3)$

Cases. $q_1 \wedge q_2$
 $q_1 \wedge q_3$
 $q_2 \wedge q_3$
 $q_1 \wedge q_2 \wedge q_3$

While it is not wrong to write the disjunction of all 4 cases, we do not need to write the 4th case. If any 2 of the members play guitar, it will already be satisfied.
 $(q_1 \wedge q_2) \vee (q_1 \wedge q_3) \vee (q_2 \wedge q_3) \vee (q_1 \wedge q_2 \wedge q_3)$
 $\equiv (q_1 \wedge q_2) \vee (q_1 \wedge q_3) \vee (q_2 \wedge q_3)$

- d) Since $\vee$ means inclusive disjunction, writing ' $p_1 \vee p_2 \vee p_3$ ' will also be true when there is more than (not exactly) one lead singer.
 So we need to specify that the other 2 are not lead singer.
 Note: usually we don't use the symbol ' $\oplus$ ' in propositional logic.